

Postulates of Quantum mechanics

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Contents

1	Postulates of Quantum Mechanics	2
1.1	Postulate 1: The State Space Postulate	2
1.2	Postulate 2: Time Evolution of Quantum States	3
1.3	Postulate 3: Measurement in Quantum Mechanics	4
1.3.1	Distinguishing Quantum States	5
1.3.2	Projective Measurements (Von Neumann Measurements)	7
1.3.3	Generalized Measurements	9
1.4	Postulate 4: Composite Quantum Systems	11

1 Postulates of Quantum Mechanics

1.1 Postulate 1: The State Space Postulate

Postulate 1 — State Space

Associated to any isolated physical system is a complex vector space with inner product, known as a Hilbert space, which defines the state space of the system. The state of the system is represented by a unit vector (a vector of norm 1) in this Hilbert space. This unit vector, also called the state vector, fully describes the system.

Explanation

- In quantum mechanics, the state of a physical system (such as a particle, atom, or qubit) is not described by classical variables like position or velocity, but rather by a vector in a complex vector space.
- This space is called a Hilbert space, which means:
 - It is a vector space over the complex numbers.
 - It has an inner product $\langle\psi|\phi\rangle$, which allows us to compute angles, lengths, and projections between vectors.
 - It is complete with respect to the norm induced by the inner product.
- The state vector $|\psi\rangle$ that represents the system must be normalized, i.e.,

$$\langle\psi|\psi\rangle = 1$$

so that the total probability of all possible measurement outcomes is 1.

- The state vector contains all physically accessible information about the system. Different vectors that differ only by a global phase (i.e., $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$) represent the same physical state.
- For our utility in this course, we just need to know what a qubit is. It has been covered earlier. For a single qubit, the state space is a 2-dimensional Hilbert space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \text{with } \alpha, \beta \in \mathbb{C} \text{ and } |\alpha|^2 + |\beta|^2 = 1$$

Here, $|0\rangle$ and $|1\rangle$ form an orthonormal basis of the Hilbert space $\mathbb{C}^2$, and α, β are complex amplitudes.

1.2 Postulate 2: Time Evolution of Quantum States

Postulate 2 — Unitary Evolution

The evolution of a closed quantum system over time is described by a unitary transformation. That is, if the system is in state $|\psi\rangle$ at time t_1 , then at a later time t_2 , the state is given by:

$$|\psi'\rangle = U |\psi\rangle$$

where U is a unitary operator.

Explanation

- When a quantum system is isolated from its environment, it evolves smoothly over time. This time evolution is always governed by a *unitary operator*.
- A unitary operator U satisfies the condition:

$$U^\dagger U = U U^\dagger = I,$$

meaning it preserves the norm (length) of state vectors. As a result, the total probability—determined by the square of the norm—remains 1 at all times.

- This property of unitary operators ensures that quantum evolution is *reversible*. If we apply U to evolve the system forward in time, we can always apply $U^\dagger$ (the conjugate transpose) to reverse the evolution.
- In contrast, classical logic gates like the NOT gate flip a bit from 0 to 1 or 1 to 0, but they are generally irreversible—we cannot recover the input just from the output without additional information.
- The quantum version of the NOT gate is implemented by the Pauli- X operator. This operator is both reversible and unitary. For example:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle$$

Since $X|0\rangle = |1\rangle$. Applying X gate again will give the input state back

$$XX|0\rangle = X(X|0\rangle) = X|1\rangle = |0\rangle \tag{1}$$

Hence, $X^2 = I$, the gate undoes itself (reversible). And, $X^\dagger = X$ as the gates are unitary. Similarly $Y^2 = Z^2 = I$.

- In general, a closed quantum system that starts in a state $|\psi(t_1)\rangle$ will evolve to a state $|\psi(t_2)\rangle$ at a later time t_2 by the application of a unitary operator $U(t_2, t_1)$:

$$|\psi(t_2)\rangle = U(t_2, t_1) |\psi(t_1)\rangle$$

This is the quantum analogue of classical time evolution laws—except we use linear operators instead of differential equations directly.

- Many quantum gates used in quantum computing are unitary. For example:
 - The Pauli- X matrix flips basis states like a classical NOT gate. It is also called a *bit-flip*.
 - The Pauli- Z matrix flips the phase of $|1\rangle$:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle$$

This is known as a *phase-flip*.

- Other important single-qubit gates like the Hadamard gate H and various rotation gates (to be discussed later) are also unitary and respect this postulate.
- Therefore, the evolution of a quantum system over time does not introduce any randomness or loss of information. It corresponds to a smooth, deterministic transformation—a rotation—of the state vector within the system's Hilbert space.

1.3 Postulate 3: Measurement in Quantum Mechanics

Postulate 3 — Measurement Postulate

When we measure a physical quantity (called an observable) in quantum mechanics, the possible results are the eigenvalues of the operator A corresponding to that observable.

Let A be a Hermitian operator representing an observable, and let its spectral decomposition be:

$$A = \sum_n a_n P_n,$$

where:

- a_n are the eigenvalues (the possible outcomes of measurement),
- $P_n = |u_n\rangle\langle u_n|$ are projection operators onto the eigenvectors $|u_n\rangle$ of A .

Probability of a Measurement Outcome

If the system is in state $|\psi\rangle$, the probability of obtaining outcome a_n when measuring A is:

$$P(a_n) = \langle\psi| P_n |\psi\rangle = \text{Tr}(P_n |\psi\rangle\langle\psi|)$$

Alternatively, if we define the probability amplitude c_n of obtaining measurement result a_n as:

$$c_n = \langle u_n | \psi \rangle,$$

then the probability becomes:

$$P(a_n) = \frac{|c_n|^2}{\langle\psi|\psi\rangle} = \frac{|\langle u_n | \psi \rangle|^2}{\langle\psi|\psi\rangle}$$

If the state $|\psi\rangle$ is normalized (i.e., $\langle\psi|\psi\rangle = 1$), then:

$$P(a_n) = |\langle u_n | \psi \rangle|^2$$

Post-Measurement State (Wavefunction Collapse)

After a measurement yielding the result a_n , the state of the system collapses into the corresponding eigenvector $|u_n\rangle$.

More generally, if $P_n = |u_n\rangle\langle u_n|$ is the projection operator onto the eigenspace of a_n , then the post-measurement state becomes:

$$|\psi\rangle \xrightarrow{\text{measurement}} \frac{P_n |\psi\rangle}{\sqrt{\langle\psi| P_n |\psi\rangle}}$$

This ensures that the resulting state is properly normalized.

The qubit is in the state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Then the probabilities of the two outcomes are:

- Probability of outcome 0:

$$P(0) = \langle\psi| P_0 |\psi\rangle = \langle\psi| 0\rangle\langle 0|\psi\rangle = |\langle 0|\psi\rangle|^2 = |\alpha|^2$$

- Probability of outcome 1:

$$P(1) = \langle\psi| P_1 |\psi\rangle = \langle\psi| 1\rangle\langle 1|\psi\rangle = |\langle 1|\psi\rangle|^2 = |\beta|^2$$

If the measurement result is 0, the state collapses to $|0\rangle$; if the result is 1, it collapses to $|1\rangle$.

We can also write the post-measurement state more formally as:

$$\begin{aligned} |\psi\rangle &\xrightarrow{\text{result } 0} \frac{P_0 |\psi\rangle}{\sqrt{\langle\psi| P_0 |\psi\rangle}} = \frac{\alpha |0\rangle}{\sqrt{|\alpha|^2}} = \frac{\alpha |0\rangle}{|\alpha|} \\ |\psi\rangle &\xrightarrow{\text{result } 1} \frac{P_1 |\psi\rangle}{\sqrt{\langle\psi| P_1 |\psi\rangle}} = \frac{\beta |1\rangle}{\sqrt{|\beta|^2}} = \frac{\beta |1\rangle}{|\beta|} \end{aligned}$$

1.3.1 Distinguishing Quantum States

One of the most important applications of Postulate 3 is to understand when quantum states can or cannot be distinguished by measurement.

Orthonormal States Can Be Perfectly Distinguished

Let's consider a scenario where Alice and Bob play a quantum guessing game:

- Alice prepares a quantum state $|\psi_j\rangle$ from a known set $\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle\}$ and sends it to Bob. The states $\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle\}$ are orthonormal.

- Bob's task is to identify which state $|\psi_i\rangle$ he received, by performing a measurement by using the measurement operators:

$$M_i = |\psi_i\rangle \langle \psi_i|$$

for each i .

These operators satisfy the completeness relation:

$$\sum_i M_i = I.$$

If Alice sends the state $|\psi_j\rangle$, then the probability of Bob correctly identifying it is:

$$P(1) = \langle \psi_j | M_1 | \psi_j \rangle = \langle \psi_j | \psi_1 \rangle \langle \psi_1 | \psi_j \rangle = 0.$$

$$P(2) = \langle \psi_j | M_2 | \psi_j \rangle = \langle \psi_j | \psi_2 \rangle \langle \psi_2 | \psi_j \rangle = 0.$$

$\vdots$

$$P(j) = \langle \psi_j | M_j | \psi_j \rangle = \langle \psi_j | \psi_j \rangle \langle \psi_j | \psi_j \rangle = 1.$$

$\vdots$

$$P(n) = \langle \psi_j | M_n | \psi_j \rangle = \langle \psi_j | \psi_n \rangle \langle \psi_n | \psi_j \rangle = 0.$$

This result means that Bob will always obtain the correct outcome j with probability 1 and 0 for the incorrect outcomes. Total probability is

$$P = \sum_{i=1}^n P(i) = P(1) + P(2) + \dots + P(j) + \dots + P(n) = 1$$

Therefore, orthonormal quantum states can always be perfectly distinguished.

Non-Orthogonal States Cannot Be Perfectly Distinguished

Now suppose Alice chooses between two quantum states $|\psi_1\rangle$ and $|\psi_2\rangle$ that are not orthogonal. In this case, we will prove that no quantum measurement can perfectly distinguish them.

Here's the intuition: Since $|\psi_2\rangle$ has a nonzero component along $|\psi_1\rangle$, any measurement that tries to distinguish $|\psi_1\rangle$ with certainty will have some chance of mistakenly identifying $|\psi_2\rangle$ as $|\psi_1\rangle$.

Example: Let's consider a set of three quantum states:

$$\left\{ |\psi_1\rangle = |0\rangle, \quad |\psi_2\rangle = |1\rangle, \quad |\psi_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right\}$$

- The states $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal, since:

$$\langle 0 | 1 \rangle = 0$$

Therefore, these two states can be perfectly distinguished by a measurement in the computational basis $\{|0\rangle, |1\rangle\}$.

- Now consider the third state, $|\psi_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. This state is not orthogonal to either $|0\rangle$ or $|1\rangle$:

$$\begin{aligned}\langle 0|\psi_3\rangle &= \frac{1}{\sqrt{2}}(\langle 0|0\rangle + \langle 0|1\rangle) = \frac{1}{\sqrt{2}}(1 + 0) = \frac{1}{\sqrt{2}} \\ \langle 1|\psi_3\rangle &= \frac{1}{\sqrt{2}}(\langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{\sqrt{2}}(0 + 1) = \frac{1}{\sqrt{2}}\end{aligned}$$

- This means that if Alice prepares $|\psi_3\rangle$ and Bob tries to measure in the computational basis, he will get:

$$P(0) = |\langle 0|\psi_3\rangle|^2 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}, \quad P(1) = \frac{1}{2}$$

So Bob can't tell whether the state was $|0\rangle$, $|1\rangle$, or $|\psi_3\rangle$ just from this measurement outcome — there is unavoidable ambiguity.

- Although $|0\rangle$ and $|1\rangle$ are distinguishable, the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ is non-orthogonal to both and cannot be perfectly distinguished from them using any quantum measurement.

1.3.2 Projective Measurements (Von Neumann Measurements)

Projective measurements are the most basic and traditional kind of quantum measurement. They are also known as *Von Neumann measurements*, named after the mathematician who first formalized them.

The idea behind a projective measurement is simple: we are given a system prepared in a state $|\psi\rangle$ and want to check whether it lies in one of several *mutually exclusive* possible states.

Properties of Projection Operators

A projection operator P satisfies the following two properties:

$$\begin{aligned}P &= P^\dagger \quad (\text{Hermitian}) \\ P^2 &= P \quad (\text{Idempotent})\end{aligned}$$

If P_1 and P_2 are two different projection operators corresponding to mutually exclusive outcomes, then they are orthogonal:

$$P_1 P_2 = 0$$

A set of projectors $\{P_i\}$ is said to be *complete* if it satisfies:

$$\sum_i P_i = I$$

This guarantees that probabilities always sum to 1.

A projective measurement is a special type of quantum measurement described by a Hermitian operator M , known as an *observable*, acting on the state space of the quantum system. The operator M admits a spectral decomposition of the form:

$$M = \sum_m m P_m$$

Here:

- m are the eigenvalues of M , which represent the possible measurement outcomes.
- $P_m = |u_m\rangle \langle u_m|$ is the projector onto the eigenspace corresponding to the eigenvalue m .

Measurement Probabilities

If the system is in state $|\psi\rangle$, the probability of obtaining outcome m when measuring M is given by:

$$P(m) = \langle \psi | P_m | \psi \rangle$$

This expression gives the probability of finding the system in the eigenspace corresponding to eigenvalue m .

Post-Measurement State (Collapse)

If the outcome m is obtained, the quantum state collapses to the corresponding eigenstate (or eigenspace). The post-measurement state is given by:

$$|\psi_{\text{after}}\rangle = \frac{P_m |\psi\rangle}{\sqrt{p(m)}} = \frac{P_m |\psi\rangle}{\sqrt{\langle \psi | P_m | \psi \rangle}}$$

This ensures that the new state is normalized.

Expectation Value of an Observable

The average (expected) value of the observable M in the state $|\psi\rangle$ is given by:

$$\langle M \rangle = \sum_m m p(m) = \sum_m m \langle \psi | P_m | \psi \rangle = \langle \psi | M | \psi \rangle$$

This formula is extremely useful and allows easy computation of expected values without summing explicitly over outcomes if we know the matrix representation of M .

Standard Deviation

The standard deviation of measurement results (i.e., how much outcomes typically deviate from the average value) is:

$$\Delta(M) = \sqrt{\langle M^2 \rangle - \langle M \rangle^2}$$

Here:

- $\langle M \rangle = \langle \psi | M | \psi \rangle$ is the average value.
- $\langle M^2 \rangle = \langle \psi | M^2 | \psi \rangle$ is the expectation of the square of the observable.

Example: Measuring a Qubit in the Computational Basis

Suppose a qubit is in the state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

We perform a measurement in the standard computational basis $\{|0\rangle, |1\rangle\}$. The corresponding projective measurement operators are:

$$P_0 = |0\rangle\langle 0|, \quad P_1 = |1\rangle\langle 1|$$

Measurement Probabilities Using the Born rule, we compute the probabilities of each outcome:

$$\begin{aligned} P(0) &= \langle\psi| P_0 |\psi\rangle = \langle\psi| |0\rangle\langle 0| |\psi\rangle = |\langle 0|\psi\rangle|^2 = |\alpha|^2 \\ P(1) &= \langle\psi| P_1 |\psi\rangle = \langle\psi| |1\rangle\langle 1| |\psi\rangle = |\langle 1|\psi\rangle|^2 = |\beta|^2 \end{aligned}$$

Post-Measurement State If the measurement result is 0, the state collapses to:

$$|\psi_{\text{after}}\rangle = \frac{P_0 |\psi\rangle}{\sqrt{P(0)}} = \frac{|0\rangle\langle 0| (\alpha |0\rangle + \beta |1\rangle)}{\sqrt{|\alpha|^2}} = |0\rangle$$

If the measurement result is 1, the state collapses to:

$$|\psi_{\text{after}}\rangle = \frac{P_1 |\psi\rangle}{\sqrt{P(1)}} = \frac{|1\rangle\langle 1| (\alpha |0\rangle + \beta |1\rangle)}{\sqrt{|\beta|^2}} = |1\rangle$$

1.3.3 Generalized Measurements

In quantum mechanics, measurements can be described in a more general way than just using projectors. This framework is called **generalized measurements** or Positive Operator-Valued Measures (POVMs).

Measurement Operators

A generalized measurement is described by a set of measurement operators $\{M_m\}$, where the index m labels the possible outcomes of the measurement. These operators act on the state space of the system and satisfy the completeness relation:

$$\sum_m M_m^\dagger M_m = I$$

This condition ensures that the total probability of all possible outcomes sums to 1.

Measurement on a State

Suppose the quantum system is in the state $|\psi\rangle$. The probability of obtaining outcome m is given by:

$$P(m) = \langle\psi| M_m^\dagger M_m |\psi\rangle$$

After the measurement, if outcome m is obtained, the state collapses to:

$$|\psi'\rangle = \frac{M_m |\psi\rangle}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$$

Special Case: Projective Measurements

Projective measurements are a special case of generalized measurements in which each M_m is a Hermitian projection operator:

$$M_m = P_m = |u_m\rangle \langle u_m|, \quad P_m^2 = P_m, \quad P_m^\dagger = P_m$$

In this case, the probability becomes:

$$\Pr(m) = \langle\psi| P_m |\psi\rangle$$

and the post-measurement state is:

$$|\psi'\rangle = \frac{P_m |\psi\rangle}{\sqrt{\langle\psi| P_m |\psi\rangle}}$$

Special Case: Projective Measurements

Projective measurements (discussed earlier) are a special case of POVMs. In that case:

- Each measurement operator is a projector: $M_m = P_m = |u_m\rangle \langle u_m|$
- Then the POVM element becomes:

$$E_m = M_m^\dagger M_m = P_m^\dagger P_m = P_m^2 = P_m$$

- The completeness relation becomes: $\sum_m P_m = I$

Therefore, in projective measurements, the POVM elements $\{E_m\}$ are orthogonal projection operators. But in general, POVMs allow E_m to be non-projective and non-orthogonal, providing more flexibility in describing realistic measurements.

1.4 Postulate 4: Composite Quantum Systems

Postulate 4 — Composite Systems

The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$.

In quantum mechanics, we do not always deal with single isolated particles. In many scenarios, particularly in quantum information processing, it becomes necessary to study *multi-particle systems*.

To describe such systems, we must mathematically construct a Hilbert space $\mathcal{H}$ that is a composite of the individual Hilbert spaces associated with each particle. The mathematical operation that facilitates this construction is called the tensor product.

Two-Particle Systems and the Tensor Product

Suppose $\mathcal{H}_1$ and $\mathcal{H}_2$ are Hilbert spaces of dimension N_1 and N_2 , respectively. We define the composite Hilbert space as:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$$

The dimension of this composite space is the product of the dimensions of the component spaces:

$$\dim(\mathcal{H}) = \dim(\mathcal{H}_1) \cdot \dim(\mathcal{H}_2) = N_1 N_2$$

Composite States in Dirac Notation

Let $|\phi\rangle \in \mathcal{H}_1$ and $|\chi\rangle \in \mathcal{H}_2$ be two state vectors. Then the state vector $|\psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ is given by their tensor product:

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$$

We often simplify notation by omitting the tensor product symbol:

$$|\phi\rangle \otimes |\chi\rangle \equiv |\phi\rangle |\chi\rangle \equiv |\phi\chi\rangle$$

Properties of the Tensor Product

The tensor product is linear in both arguments. For vectors:

$$\begin{aligned} |\phi\rangle \otimes (|\chi_1\rangle + |\chi_2\rangle) &= |\phi\rangle \otimes |\chi_1\rangle + |\phi\rangle \otimes |\chi_2\rangle \\ (|\phi_1\rangle + |\phi_2\rangle) \otimes |\chi\rangle &= |\phi_1\rangle \otimes |\chi\rangle + |\phi_2\rangle \otimes |\chi\rangle \end{aligned}$$

The tensor product is also compatible with scalar multiplication:

$$|\phi\rangle \otimes (\alpha |\chi\rangle) = \alpha (|\phi\rangle \otimes |\chi\rangle) = (\alpha |\phi\rangle) \otimes |\chi\rangle$$

Basis for Composite Systems

If $\{|u_i\rangle\}$ is a basis for $\mathcal{H}_1$ and $\{|v_j\rangle\}$ is a basis for $\mathcal{H}_2$, then a basis for the composite space $\mathcal{H}_1 \otimes \mathcal{H}_2$ is given by the set:

$$\{|w_{ij}\rangle = |u_i\rangle \otimes |v_j\rangle\}$$

This construction implies that for every pair (i, j) , there is a corresponding basis vector in the composite space.

Commutativity of Tensor Products

While mathematically the tensor product is not commutative (i.e., $|\phi\rangle \otimes |\chi\rangle \neq |\chi\rangle \otimes |\phi\rangle$ in general), in many physical contexts where particles are indistinguishable or where labeling is arbitrary, one may informally write:

$$|\phi\rangle \otimes |\chi\rangle = |\chi\rangle \otimes |\phi\rangle$$

However, care must be taken with this symmetry assumption, especially in quantum computing where tensor product ordering corresponds to qubit register indexing. This exponential growth of the state space underlies both the power and the complexity of quantum computing.